

Programming Language Foundations

Appunti

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Contents

Chapter 1

Introduction and Mathematical Preliminaries	Page
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Introduction and Mathematical Preliminaries

Binary relations, functions (total/partial), operational semantics
Derivation tree of an arithmetic expression using the inference rules. (slide 2)

Definition 1.0.1: Multi-step evaluation

Binary relation indicated with $\rightarrow^*$ which is the reflexive and transitive closure of the small-step evaluation.

Note that the transitive rule is a single step and a multi step, although it can be proved that it allows concatenation of single steps.

Properties of programs

- Determinism: suppose that for any given configuration, if this can step in different ways these are equal
- Termination: evaluation terminates. Formally, for the multi step there is a configuration that it can multi step to without other possible steps.
- Soundness: if we run a program long enough we should get a final integer, but it's not true because stores are partial function and if a variable isn't defined we're cooked.

Definition 1.0.2: Free Variables

Given the function $fvs(e)$ defined inductively as:

Note that in $x := x + 1; y := x$ we still need to add x to the free variables because it's in the rhs of the first assignment.

Definition 1.0.3: Well-Formedness

Example of an early type system. All the variables that e refers to have values in σ .

Two properties that imply soundness:

Lemma 1.0.1 Progress

Lemma 1.0.2 Preservation

with these lemmas and termination we can show that a well-formed configuration is sound.
A bit of maths before the proof

Definition 1.0.4: Inductive Set

We have used inductive sets with the inductive rules (remember that $\rightarrow$ is a relation, so a set of tuples) and BNF grammar, free-variables and multistep. Another example is the natural numbers using the successor definition. If we want to prove something about an inductive set (that can be infinite by closing under the inductive rules) we can use the induction principle.

The base cases are all the axioms.

Sometimes the properties are defined over multiple inductive sets, so we need to choose one.